

Python Dictionary Practice Problems

A comprehensive workbook featuring basic operations, nested manipulation, and intermediate problem-solving.

Part 1: Basic Operations & Syntax

Problem 1: Create and Access

Create a dictionary with keys 'brand', 'model', and 'year'. Write a code snippet to access the value of 'model' safely using two different approaches.

```
Expected Output:  
Approach 1: dict['model']  
Approach 2: dict.get('model')
```

Problem 2: Add and Update Elements

Given an initial dictionary {'a': 10, 'b': 20}, add a new key-value pair 'c': 30 and update the value of 'b' to 25.

```
Input:  
d = {'a': 10, 'b': 20}  
Expected Output:  
{'a': 10, 'b': 25, 'c': 30}
```

Problem 3: Safe Deletion

Write a code snippet to remove the key 'salary' from a dictionary if it exists, without raising a KeyError if it's missing.

```
Input:  
d = {'name': 'Alice', 'age': 30}  
Expected Output:  
{'name': 'Alice', 'age': 30}
```

Part 2: Iteration & Transformation

Problem 4: Two Lists to Dictionary

Given two lists, one containing keys and another containing values, map them into a single dictionary using the zip() function.

```
Input:  
keys = ['Ten', 'Twenty', 'Thirty']
```

```
values = [10, 20, 30]
Expected Output:
{'Ten': 10, 'Twenty': 20, 'Thirty': 30}
```

Problem 5: Frequency Counter

Write a Python function that takes a list of words and returns a dictionary detailing the frequency count of each unique word.

```
Input:
words = ['apple', 'banana', 'apple', 'orange', 'banana', 'apple']
Expected Output:
{'apple': 3, 'banana': 2, 'orange': 1}
```

Problem 6: Dictionary Comprehension

Use a dictionary comprehension to generate a dictionary where the keys are numbers from 1 to 5 and the values are the squares of those numbers.

```
Expected Output:
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

Part 3: Intermediate Problems & Merging

Problem 7: Merge with Summed Conflict Values

Write a function to merge two dictionaries. If a key exists in both warehouses, sum their inventory values together.

```
Input:
w1 = {'apples': 30, 'bananas': 15}
w2 = {'oranges': 25, 'apples': 20}
Expected Output:
{'apples': 50, 'bananas': 15, 'oranges': 25}
```

Problem 8: Filter Dictionary

Filter a dictionary of student grades keeping only the entries where the score is strictly greater than 80.

```
Input:
scores = {'Alice': 85, 'Bob': 72, 'Charlie': 90}
Expected Output:
{'Alice': 85, 'Charlie': 90}
```

Problem 9: Nested Dictionary Access

Given a nested employee dictionary, change the value of the nested key 'salary' inside the 'manager' section to 8500.

Input:

```
emp = {'id': 1, 'role': {'manager': {'name': 'Brad', 'salary': 500}}}
```

Expected Output:

```
{'id': 1, 'role': {'manager': {'name': 'Brad', 'salary': 8500}}}
```